

Talha Khan

AI/ML Engineer — Computer Vision & Edge AI

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SUMMARY

AI/ML engineer with 4+ years building computer vision, edge AI, and predictive maintenance systems that run in real production environments — from a fuel-cell manufacturing line to edge devices wired directly into industrial PLCs. Experienced leading AI/ML and data engineering initiatives end-to-end: data pipeline, model development, edge deployment, and production monitoring. Focused on systems that hold up under real hardware constraints and messy production data, not just notebook accuracy.

CORE SKILLS

Languages	Python, MATLAB, SQL
Deep Learning	TensorFlow, PyTorch, Keras
Computer Vision	OpenCV, Picamera2, VGG-19 CNN, Image Processing
Classical ML	XGBoost, Scikit-learn, SciPy
Data Engineering	PyArrow, Pandas, Dask, Apache Spark
Edge & Robotics	Raspberry Pi, ESP32, Jetson Xavier/Nano, PLC / TCP-IP
Infrastructure	Docker, AWS, RHEL, PostgreSQL
Reporting & MLOps	Power BI, MLflow, Streamlit, Microsoft Teams

SELECTED ENGINEERING WORK

Automated Chassis Serial-Number Recognition System

Automotive Manufacturing / Industrial Computer Vision

Built an edge-deployed VGG-19 CNN reading robot-written chassis serial numbers on Raspberry Pi 4B, quantized to INT8 to fit inside a 3-second robot cycle, signaling pass/fail directly to a TOYOPUC PLC. Eliminated 320 man-hours/month of manual verification.

In-House Vision-Guided Trolley AGV

Automotive Manufacturing / Robotics & Industrial AI

Designed and built an autonomous guided vehicle from scratch in Python — 60 FPS OpenCV tape detection, PID-based wheel control, ESP32/Raspberry Pi hardware — retrofitted onto standard production trolleys. Eliminated 96 man-hours/month and won a company-wide excellence award.

High-Volume Time-Series ETL & Parquet Pipeline

Automotive Manufacturing / Data Engineering

Built a real-time CSV-to-Parquet pipeline for 100+ PLC devices using a PostgreSQL FIFO queue and watchdog-based ingestion, cutting annual storage from ~2,300GB to ~130GB while enabling instant, real-time search across the dataset.

Early Anomaly Prediction for Servo Motors

Automotive Manufacturing / Predictive Maintenance

Built an XGBoost-based anomaly-prediction system across 500+ servo motors using a generalistic cycle-cutting algorithm and statistical feature vectors, with results delivered via Power BI and Microsoft Teams. Improved early

detection by 86% and cut production stoppage by 20 hours/month.

PROFESSIONAL EXPERIENCE

Team Lead — Data Engineering & AI/ML

Fuel-Cell Manufacturing Plant, Japan — Jan 2023 to Present

Leading on-prem and cloud data engineering for a next-generation fuel-cell manufacturing line, including system, database, application, and network architecture. Decreased data processing time by up to 60% and compressed time-series production data by up to 93%. Delivered the edge computer-vision, robotics, ETL, and predictive-maintenance systems above.

AI Consultant — Machine Learning / Deep Learning

AI Infinity, Japan — Apr 2022 to Sep 2022

Deployed an NLP-based application on AWS and automated the backend AI pipeline in Python; migrated NLP products to AWS, cutting on-prem cost by roughly 70%.

AI Engineer — Machine Learning / Deep Learning

NEC Platforms Ltd, Japan — Oct 2021 to Mar 2022

Trained and evaluated AI models on primarily image-based data; developed image-processing algorithms (LAB color space) that increased accuracy by 40%, and improved a custom FCN model structure for a 20% accuracy gain.

ENGINEERING PHILOSOPHY

Models built for the constraints they'll actually run under — edge hardware, cycle time, real shop-floor conditions — validated with domain experts, monitored and reported through tools teams already use, and designed to keep working long after the pilot ends.